

Continual and Data-Efficient Reinforcement Learning for Language Model Reasoning

A Study of Replay Strategies for Sequential Mathematical and Programming Tasks

Evan Jean-Christ Zigu

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Abstract

Large language models can improve their reasoning capabilities through supervised fine-tuning and reinforcement learning, but sequential adaptation to new tasks may degrade previously acquired skills. This phenomenon is closely related to catastrophic forgetting in continual learning. This work studies continual reasoning adaptation on Qwen2.5-0.5B-Instruct using a sequence of mathematical and programming tasks derived from GSM8K, MATH, and MBPP.

The model is first adapted to applied mathematical reasoning and is then trained sequentially on advanced algebra, competition mathematics, and Python programming. Three continual-learning strategies are compared: No Replay, Fixed Replay, and Adaptive Repla. Training uses a correctness-only reward and a constant optimization budget of 40 groups with four sampled generations per group at each stage.

On the retrospective validation benchmark, Fixed Replay provides the best overall trade-off between new-task learning and retention. It reaches a final average accuracy of 26.06%, compared with 23.69% without replay and 25.03% with the current adaptive strategy. Fixed Replay also reduces average forgetting from 2.33 percentage points to 0.00 and produces positive backward transfer of +2.00 percentage points. The results therefore support replay as an effective mechanism for reducing forgetting in this small-scale continual reasoning setting. However, the current adaptive controller does not yet demonstrate a clear advantage over fixed replay.

1 Introduction

Large language models (LLMs) are increasingly expected to acquire new capabilities after their initial pre-training. In practice, this adaptation may occur through supervised fine-tuning, reinforcement learning, preference optimization, or task-specific continual training. A major difficulty appears when several capabilities are learned sequentially. Improving performance on a new task can modify model parameters in a way that degrades performance on previously learned tasks. This problem is known as *catastrophic forgetting* [5, 7].

The issue is particularly relevant for reasoning models. A model may, for example, first learn arithmetic word problems, then algebraic reasoning, then competition mathematics, and finally programming. A useful continual-learning system should acquire each new capability without unnecessarily sacrificing earlier ones. Replay is one of the simplest mechanisms for addressing this problem. Instead of training exclusively on the current task, a small fraction of examples from previously encountered domains is reintroduced during training. The model is therefore periodically reminded of earlier capabilities. This work investigates whether replay can improve continual reasoning adaptation in a small language model under a limited computational budget. The study addresses the following research question:

Can experience replay reduce catastrophic forgetting when a small language model is sequentially trained on mathematical reasoning and programming tasks?

A second question concerns replay allocation:

Can an adaptive replay mechanism provide a better plasticity–retention trade-off than a fixed replay strategy?

The experiments compare three strategies:

- **No Replay:** only the current task is used.
- **Fixed Replay:** 20% of the training groups are sampled from previously learned domains.
- **Adaptive Replay:** replay slots are allocated using estimates of previous-task performance.

The main result is that replay improves retention in the studied configuration. Fixed Replay obtains the strongest final performance and completely removes measured average forgetting on the previous reasoning domains.

2 Background

2.1 Continual Learning

Continual learning studies models that acquire knowledge from a sequence of tasks or data distributions rather than from one fixed dataset [7].

Let

$$\mathcal{T}_0, \mathcal{T}_1, \dots, \mathcal{T}_T$$

denote a sequence of tasks.

After training on task $\mathcal{T}_t$, the model parameters become

$$\theta_t.$$

The objective is not only to obtain strong performance on $\mathcal{T}_t$, but also to preserve useful knowledge acquired from tasks

$$\mathcal{T}_0, \dots, \mathcal{T}_{t-1}.$$

The central stability–plasticity trade-off is therefore:

$$\text{plasticity} \quad \leftrightarrow \quad \text{retention}.$$

High plasticity enables rapid adaptation to new tasks, while high stability preserves previous knowledge.

2.2 Catastrophic Forgetting

Catastrophic forgetting occurs when training on a new task causes a substantial decrease in performance on previously learned tasks.

If

$$A_{t,d}$$

denotes the accuracy of the model after training stage t on domain d , forgetting for domain d can be written as

$$F_d = \max_{k \geq t_d} A_{k,d} - A_{T,d},$$

where t_d is the stage at which domain d is learned and T is the final stage.

A larger value of F_d indicates stronger forgetting.

2.3 Experience Replay

Experience replay maintains a memory containing examples from previous tasks.

At training stage t , the learning dataset can be represented as

$$\mathcal{D}^{(t)} = \mathcal{D}_{\text{current}}^{(t)} \cup \mathcal{M}^{(t)},$$

where $\mathcal{M}^{(t)}$ contains replay examples.

Replay is attractive because it is simple, model-agnostic, and does not require substantial architectural changes [2].

2.4 GRPO and Verifiable Rewards

The reinforcement-learning experiments use a GRPO-style training procedure inspired by DeepSeek-Math [6].

For a prompt x , several candidate responses

$$y_1, \dots, y_G$$

are generated.

Each candidate receives a reward r_i . The relative reward inside the group is then used to construct a learning signal.

In this project, the final reward is intentionally simple:

$$r(y) = \begin{cases} 1, & \text{if the answer is correct,} \\ 0, & \text{otherwise.} \end{cases}$$

Mathematical answers are checked using deterministic answer verification, while programming outputs are evaluated through executable unit tests in a sandbox.

3 Benchmark and Task Sequence

3.1 Datasets

The benchmark combines three established reasoning datasets.

GSM8K contains grade-school mathematical word problems requiring multi-step reasoning [3].

MATH contains more difficult mathematical problems covering multiple topics and competition-style reasoning [4].

MBPP contains natural-language Python programming tasks with associated unit tests [1].

3.2 Continual Task Sequence

The continual curriculum is organized into four stages.

Table 1: Continual reasoning curriculum.

Stage	Domain	Source
0	Applied mathematics	GSM8K
1	Advanced algebra	MATH
2	Competition mathematics	MATH
3	Python programming	MBPP

Stage 0 provides the common initialization for all continual-learning strategies.

The model is first adapted using supervised fine-tuning on GSM8K. The resulting adapter is then used as the shared starting point for the three continual-learning branches.

4 Experimental Setup

4.1 Base Model

All experiments use

Qwen2.5-0.5B-Instruct.

The model is sufficiently small to perform local experiments on Apple Silicon while still supporting instruction-following and multi-step generation.

Parameter-efficient adaptation is performed using LoRA adapters.

4.2 Validation Benchmark

The retrospective validation benchmark contains 440 examples.

Table 2: Validation benchmark composition.

Domain	Examples
GSM8K / applied mathematics	150
Advanced algebra	100
Competition mathematics	100
MBPP programming	90
Total	440

The validation data are not used for parameter updates. They are evaluated retrospectively after training.

A separate independent test set containing 714 examples is kept closed throughout the experiments reported here.

4.3 Training Budget

For each continual stage, the training budget is

40 groups $\times$ 4 generations per group.

This corresponds to

$$160$$

sampled generations per training stage.

The reward is correctness-only.

4.4 No Replay

The No Replay strategy uses

$$40$$

current-task groups and no examples from previous domains.

4.5 Fixed Replay

Fixed Replay uses a replay ratio of

$$\frac{8}{40} = 20\%.$$

Each stage therefore contains

$$32$$

current-task groups and

$$8$$

replay groups.

The replay allocation is:

Table 3: Fixed replay allocation.

Stage	Replay allocation
1	8 GSM8K
2	4 GSM8K + 4 Algebra
3	3 GSM8K + 3 Algebra + 2 Competition Math

4.6 Adaptive Replay

The adaptive strategy estimates previous-domain performance using small training-only probe sets.

For domain i , the initial priority mechanism is based primarily on measured forgetting:

$$P_i = \epsilon + \max\left(0, A_i^{\text{best}} - A_i^{\text{current}}\right).$$

Replay slots are then distributed proportionally to these priorities.

In the seed-42 experiment, however, the resulting integer replay allocations were identical to the fixed replay allocation at every stage.

Consequently, the present experiment does not provide evidence that adaptive allocation itself is superior to fixed allocation.

5 Evaluation Metrics

5.1 Final Accuracy

For each domain d , the final accuracy is

$$A_{T,d},$$

where $T = 3$ is the final training stage.

5.2 Average Accuracy

The final average domain accuracy is

$$\bar{A}_T = \frac{1}{D} \sum_{d=1}^D A_{T,d}.$$

Since four domains are considered,

$$D = 4.$$

5.3 Forgetting

Forgetting on domain d is defined as

$$F_d = \max_{k \geq t_d} A_{k,d} - A_{T,d}.$$

The reported average forgetting is computed over the three domains learned before the final programming stage.

5.4 Backward Transfer

Backward transfer measures whether subsequent training improves or degrades previously learned tasks.

For domain d ,

$$BWT_d = A_{T,d} - A_{t_d,d}.$$

Positive BWT indicates that later training improved performance on the previous task.

6 Results

6.1 Final Validation Performance

Table 4 presents the final validation results.

Table 4: Final continual-learning validation results for seed 42.

Strategy	GSM8K	Algebra	Competition	MBPP	Average	Forgetting	BWT
No Replay	47.33%	15.00%	8.00%	24.44%	23.69%	2.33	-1.89
Fixed Replay	48.00%	17.00%	17.00%	22.22%	26.06%	0.00	+2.00
Adaptive Replay	46.67%	17.00%	12.00%	24.44%	25.03%	1.00	-0.44

Fixed Replay obtains the highest final average accuracy:

$$26.06\%.$$

The improvement compared with No Replay is

$$26.06 - 23.69 = \boxed{+2.37 \text{ percentage points}}.$$

Adaptive Replay reaches

$$25.03\%,$$

corresponding to

$$25.03 - 23.69 = \boxed{+1.34 \text{ percentage points}}$$

relative to No Replay.

6.2 Forgetting

The No Replay strategy exhibits an average forgetting score of

$$2.33$$

percentage points.

Adaptive Replay reduces this value to

$$1.00,$$

while Fixed Replay obtains

$$\boxed{0.00}.$$

Under the chosen metric, no average final forgetting is observed for Fixed Replay.

6.3 Backward Transfer

No Replay produces negative backward transfer:

$$BWT = -1.89.$$

Adaptive Replay also remains slightly negative:

$$BWT = -0.44.$$

In contrast, Fixed Replay produces

$$BWT = +2.00.$$

This indicates that, in this experimental run, later training not only preserves earlier abilities but slightly improves their average final performance relative to the point at which they were initially learned.

7 Visual Analysis

The following figures are generated automatically from the final validation results.

7.1 Final Average Accuracy

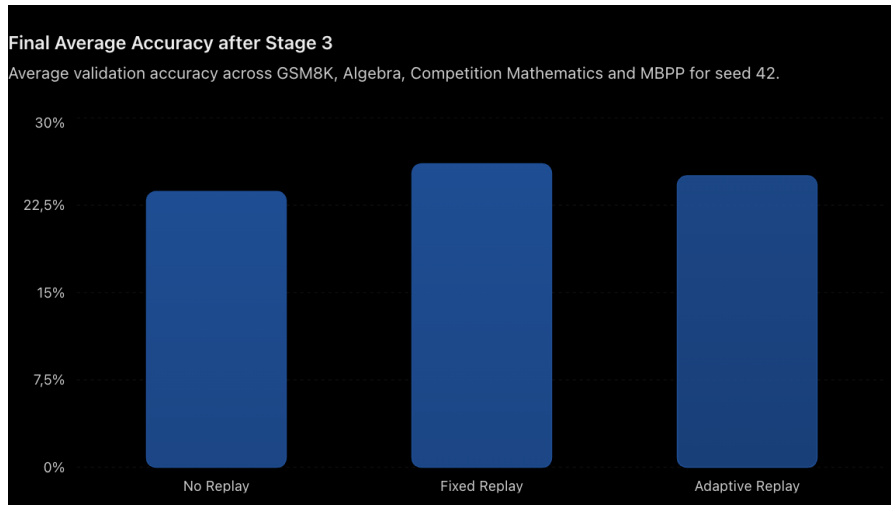


Figure 1: Final average validation accuracy after the complete continual-learning sequence.

7.2 Domain-Level Performance

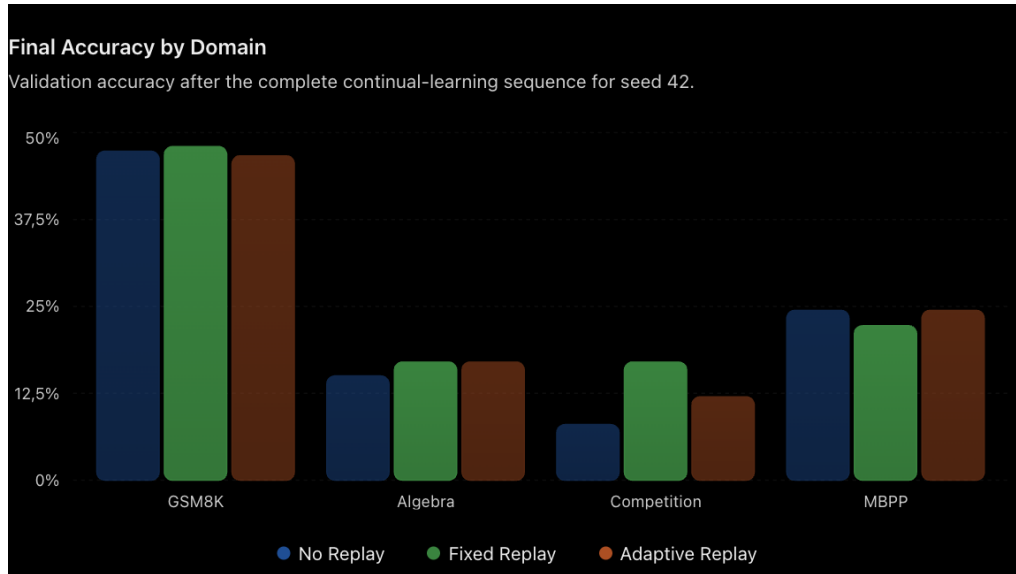


Figure 2: Final validation accuracy for each reasoning domain.

7.3 Average Forgetting

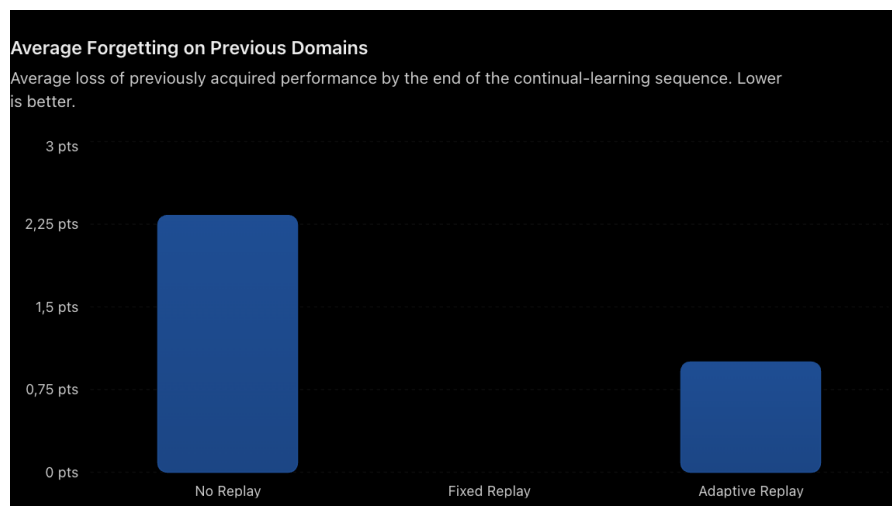


Figure 3: Average forgetting measured on previously learned domains. Lower is better.

7.4 Backward Transfer

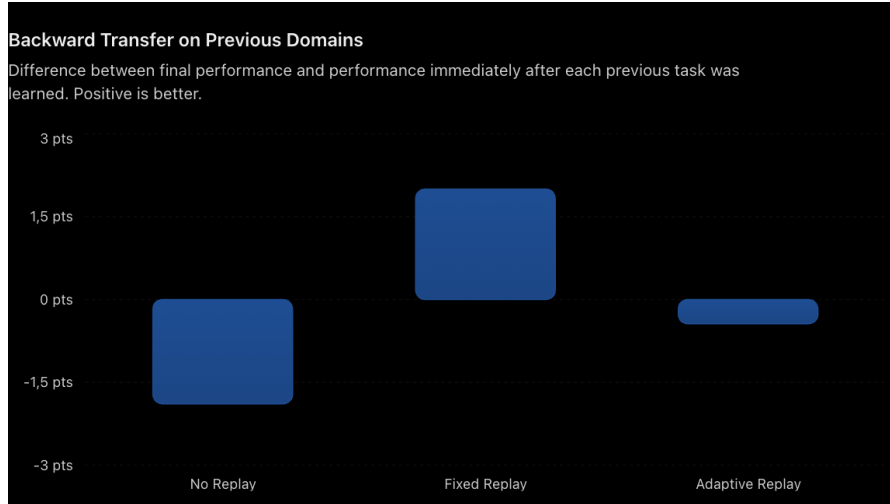


Figure 4: Average backward transfer after the complete training sequence. Positive values indicate beneficial transfer.

8 Discussion

8.1 Does Replay Reduce Forgetting?

The results support the hypothesis that replay can improve retention during continual reasoning adaptation. Compared with No Replay, both replay strategies obtain higher final average accuracy and lower forgetting. The strongest result is obtained with Fixed Replay. Its final average accuracy improves by

$$+2.37$$

percentage points relative to No Replay, while average forgetting decreases from

$$2.33$$

to

$$0.00.$$

These results suggest that even a relatively small replay budget can significantly stabilize sequential adaptation.

8.2 Plasticity–Retention Trade-off

Replay introduces an important trade-off. Because the total number of training groups remains fixed, replay strategies use fewer examples from the current task than No Replay.

For example,

$$40$$

current-task groups are used without replay, whereas replay strategies use only

current-task groups together with

8

past-task groups. Replay therefore exchanges part of the current-task training budget for better long-term retention. The programming results illustrate this trade-off.

No Replay obtains

24.44%

on MBPP, compared with

22.22%

for Fixed Replay.

However, Fixed Replay substantially improves the earlier mathematical domains and therefore achieves the highest overall average accuracy.

8.3 Why Did Adaptive Replay Not Win?

Adaptive Replay reaches

25.03%

average final accuracy and therefore outperforms No Replay.

However, it remains below Fixed Replay. More importantly, the adaptive controller produced the same stage-level integer replay allocations as Fixed Replay in this experiment. The difference between both methods therefore cannot be confidently attributed to adaptive allocation. One limitation is the small probe set used by the controller. With only eight examples per domain, probe accuracy changes in steps of

$$\frac{1}{8} = 12.5\%.$$

This produces a noisy and coarse estimate of forgetting. A more robust controller could combine several signals:

$$P_i = \alpha F_i + \beta(1 - A_i) + \gamma U_i,$$

where

- F_i represents measured forgetting,
- A_i represents current performance,
- U_i represents uncertainty or learning difficulty.

Such a controller is left for future work.

9 Limitations

Several limitations must be considered when interpreting the results.

First, the final continual-learning comparison reported in this study uses a single random seed:

$$\text{seed} = 42.$$

The results should therefore be interpreted as an experimental observation rather than as a statistically established ranking of the methods. Second, the base model contains only approximately 0.5 billion parameters. Its absolute performance on advanced algebra and competition mathematics remains limited.

Third, the adaptive replay controller relies on small probe sets and did not produce meaningfully different replay allocations from the fixed strategy in this experiment.

Fourth, replay strategies and No Replay use the same total number of training groups but not the same number of current-task examples. This design corresponds to a fixed total training budget, but the plasticity–retention trade-off should be considered when comparing new-task performance.

Finally, the independent test benchmark containing 714 examples was not evaluated during the experiments reported here. This preserves it for a future final evaluation after model-selection decisions are frozen.

10 Conclusion

This work investigated continual and data-efficient reinforcement learning for reasoning in a small language model. The model was sequentially exposed to applied mathematics, advanced algebra, competition mathematics, and Python programming.

Three strategies were compared:

$$\text{No Replay}, \quad \text{Fixed Replay}, \quad \text{Adaptive Replay}.$$

The experiments show that replay improves the retention of previous reasoning capabilities under a fixed training-group budget. Fixed Replay provides the strongest overall result, reaching a final average validation accuracy of

$$\boxed{26.06\%},$$

compared with

$$23.69\%$$

without replay.

It also reduces measured average forgetting from

$$2.33$$

percentage points to

$$\boxed{0.00}$$

and produces positive backward transfer of

+2.00

percentage points.

Adaptive Replay also improves over No Replay but does not outperform Fixed Replay in the current implementation. The main supported conclusion is therefore not that adaptive replay is superior, but that experience replay is an effective mechanism for mitigating forgetting during sequential reasoning adaptation in the studied setting. Future work should evaluate the method across multiple random seeds, improve adaptive replay allocation, investigate larger language models, and eventually evaluate the frozen configuration on the independent test benchmark.

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